Lecture 6:
Quantitative Reasoning I



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Disclaimer: this lecture references material adapted from Kosky, Balmer, Keat & Wise, Exploring Engineering (5th ed.), and from the instructor's own course slides, with modifications. No authorship of third-party material is claimed.

Last Week, in One Breath

Models — simplified representations, built for a purpose.

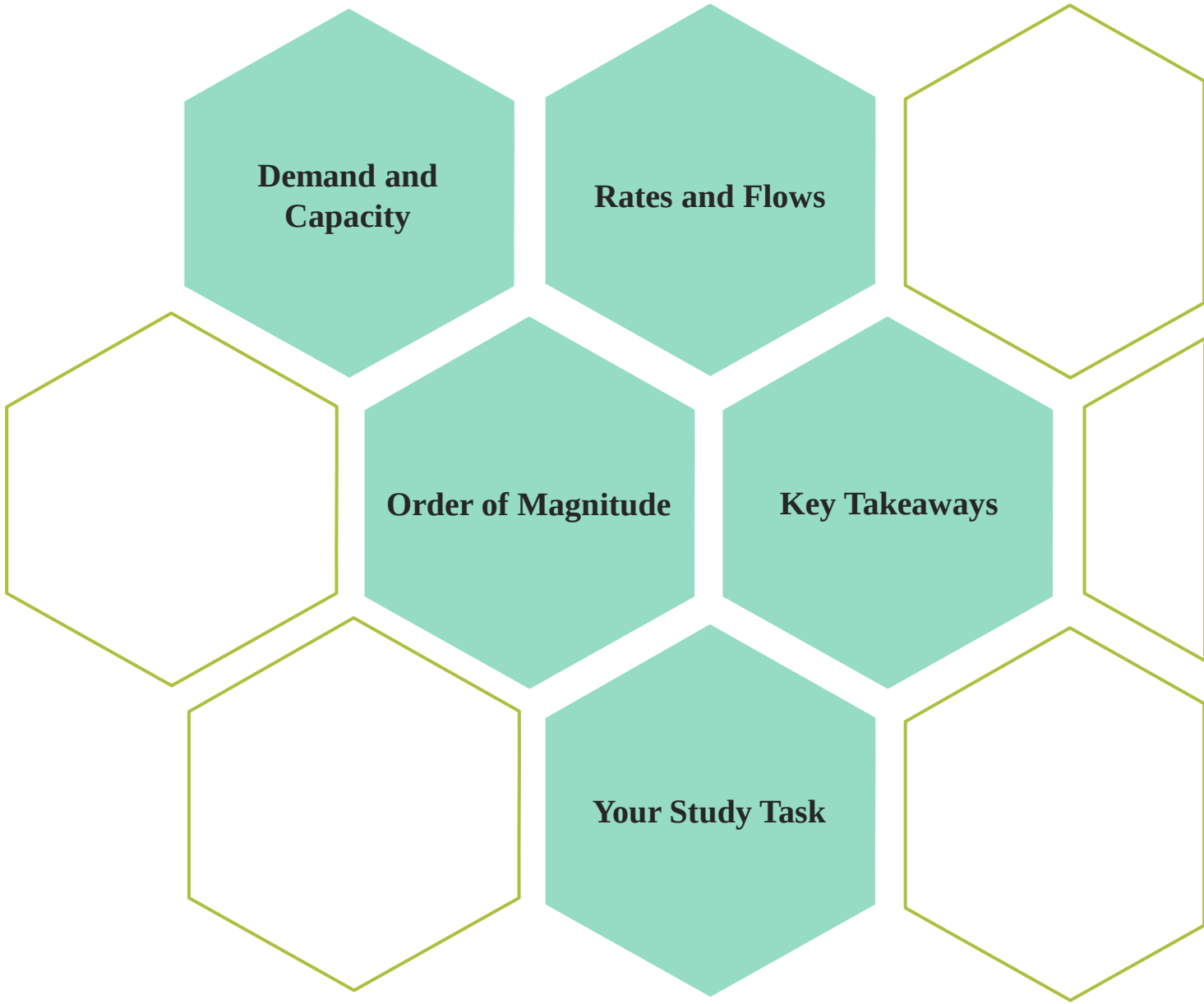
Four parts — inputs, parameters, relationships, outputs, all with units.

Assumptions — what we ignore, fix, or leave outside; we write them down.

Fit for purpose — the simplest model that answers the question well enough to decide.

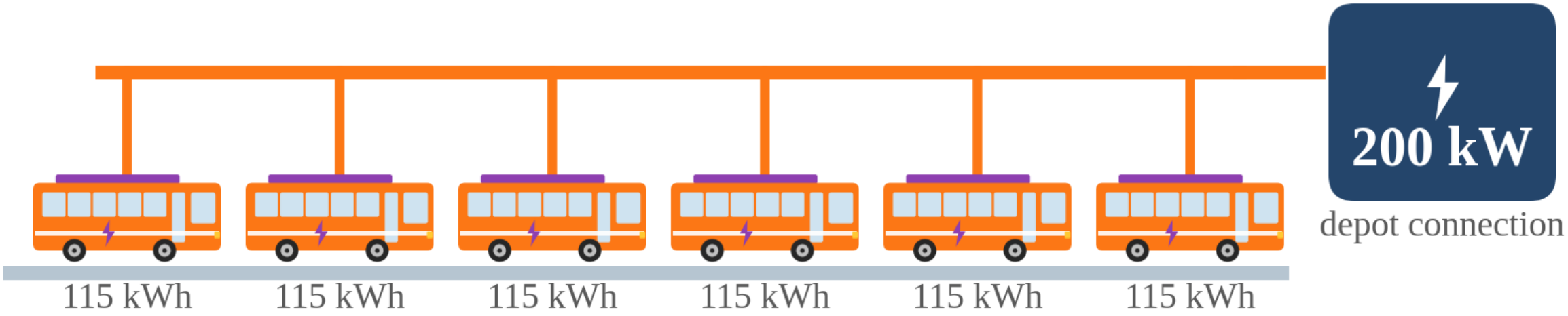
Today — put numbers on demand, capacity, rates, and flows, and check them by estimating.

Agenda


Demand and Capacity
Rates and Flows
Order of Magnitude
Key Takeaways
Your Study Task

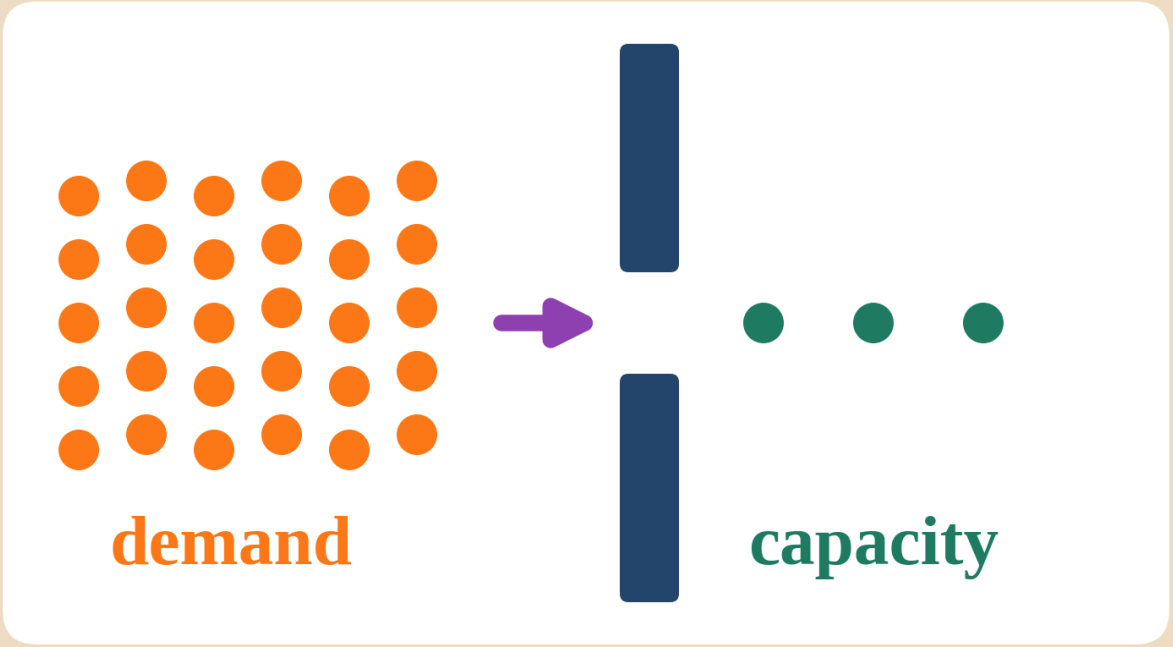
Can Six Buses Charge Overnight?

Overnight window: 22:00 to 06:00 (8 h)



Six buses, each needing 115 kWh by 06:00, and one 200 kW connection. ***Is the problem about energy (kWh) or power (kW)?***

PART 1



Demand and Capacity

What is asked of a system, and what it can deliver.

Demand and Capacity

Demand is what a system is asked to deliver. **Capacity** is the most it can deliver. Both need units and a time basis.

DEMAND

What is asked

Riders who want the 8:30 bus. kW the buses draw when plugged in.

CAPACITY

What can be delivered

Seats leaving per hour. kW the connection can supply.

TIME BASIS

Always per something

Riders per hour, kW at 13:00, kWh per day.

A number without its time basis is as incomplete as a number without units.

Idea: Kosky et al., Ch. 8 (highway capacity as a rate).

Demand and Capacity in Four Systems

System	Demand	Capacity	Unit
Campus bus at peak	300 riders want to ride	240 riders carried (40 seats × 6 buses)	per hour
Dining hall at lunch	360 diners arrive	400 meals can be served	per hour
Depot connection	360 kW if six chargers run	200 kW the connection supplies	kW
Faculty building at peak	180 kW drawn at 13:00	250 kW transformer rating	kW

Always compare like with like: same unit, same time basis.

Illustrative values, not data.

Utilisation: Demand ÷ Capacity

0.72

Faculty building

180 ÷ 250 kW

comfortable

0.90

Dining hall

360 ÷ 400 per hour

stressed

1.25

Campus bus

300 ÷ 240 per hour

over capacity

1.80

Depot, all six

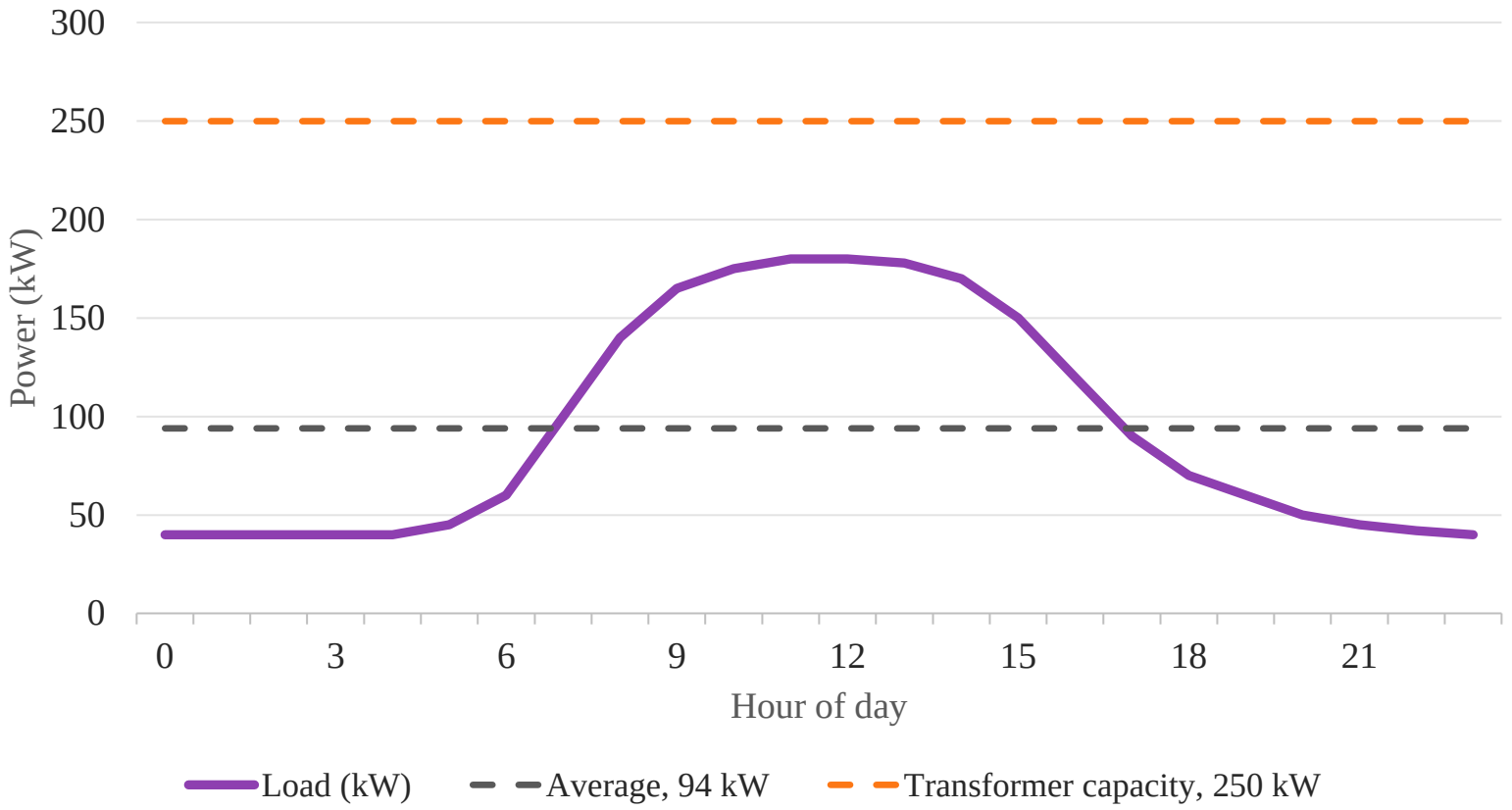
360 ÷ 200 kW

over capacity

Below about 0.8 is comfortable. Near 1 is stressed. Above 1, demand is turned away or a queue grows.

Illustrative values and rule-of-thumb thresholds.

The Average Hides the Peak



FACULTY BUILDING

Average: 94 kW

Peak: 180 kW at 13:00

Capacity: 250 kW

Sized for the average, the building would fail at midday.

Size for the peak, not for the average.

Illustrative load profile, not measured data.

Three Ways to Close a Gap

OPTION 1

Reduce or shift demand

Stagger the charging: two batches of three buses.

OPTION 2

Raise capacity

A bigger connection or transformer. It costs money and time.

OPTION 3

Accept waiting

Let a queue form, and decide how long a wait is tolerable.

Which option is best is a trade-off. We return to it in week 10.

Demand or Capacity?

In pairs: label each statement D (demand) or C (capacity). Then pair them up and compute each utilisation.

A The depot connection can supply 200 kW.

B Six buses plugged in at 22:00 would draw 360 kW.

C The bus stop can carry at most 240 riders per hour.

D 300 riders per hour want the 8:30 bus.

E The faculty transformer is rated at 250 kW.

F The faculty building peaks at 180 kW at 13:00.

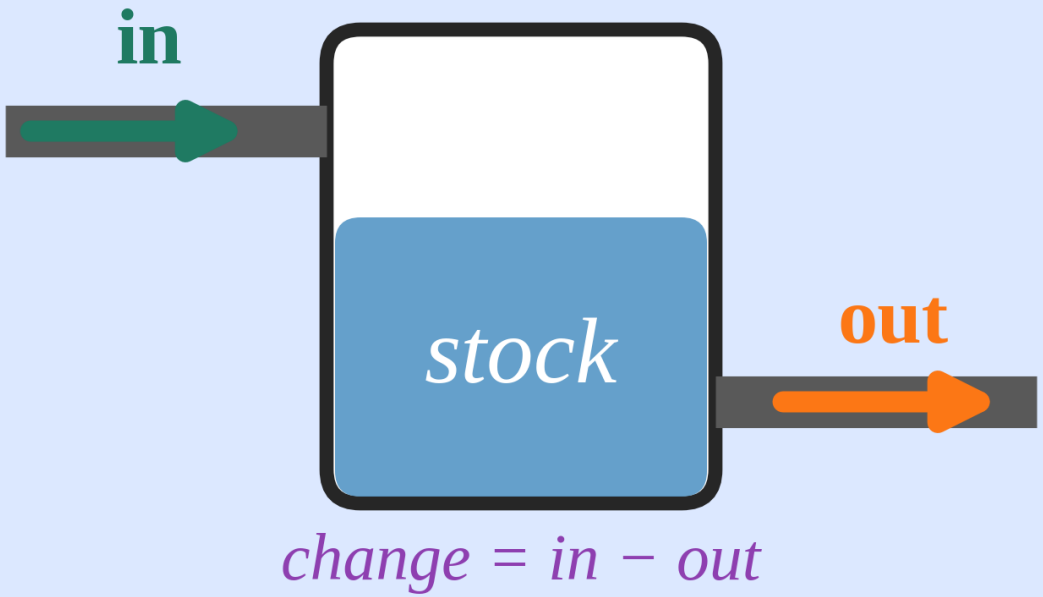
Five minutes. Watch the units and the time basis.

Demand or Capacity? Answer Key

	Kind	Pair and utilisation
A	Capacity	With B: 360 ÷ 200 = 1.8, over capacity.
B	Demand	With A: 360 ÷ 200 = 1.8, over capacity.
C	Capacity	With D: 300 ÷ 240 = 1.25, over capacity.
D	Demand	With C: 300 ÷ 240 = 1.25, over capacity.
E	Capacity	With F: 180 ÷ 250 = 0.72, comfortable.
F	Demand	With E: 180 ÷ 250 = 0.72, comfortable.

Check the unit and the time basis on both sides before you divide.

PART 2



Rates and Flows

Quantity per time, and what builds up or drains away.

A Rate Is a Quantity per Unit Time

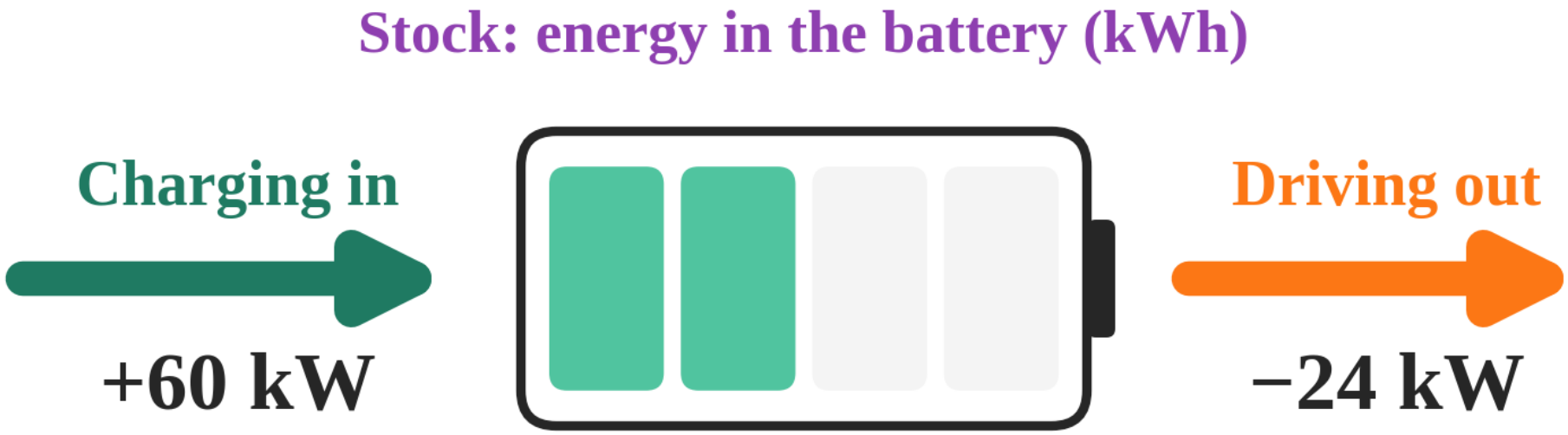
Rate	Written as	Example
Speed	km/h	The bus moves at 20 km/h.
Power	kW = kWh/h	The charger delivers 60 kW.
Arrival rate	people/min	10 students per minute reach the dining hall.
Daily energy	kWh/day	One bus uses 115 kWh per day.

kW is a rate. kWh is a quantity. Power × time = energy.

Carry units in square brackets, as Kosky et al. do: 60 [kW] × 2 [h] = 120 [kWh].

Units in square brackets: Kosky et al., Ch. 2 (Example 2.4).

Stock and Flow



Change in stock = flow in − flow out

Driving at 20 km/h: 1.2 [kWh/km] × 20 [km/h] = 24 [kW], so 160 kWh lasts about 6.7 h.

Illustrative values.

Worked Example: Charging One Bus

NEED

The variable we want

Time to restore one day's energy (h).

KNOW

What we already know

Energy 115 kWh.
Charger power 60 kW.

HOW

The model and assumptions

Time = energy ÷ power.
Constant power, no losses.

SOLVE

Do the calculation

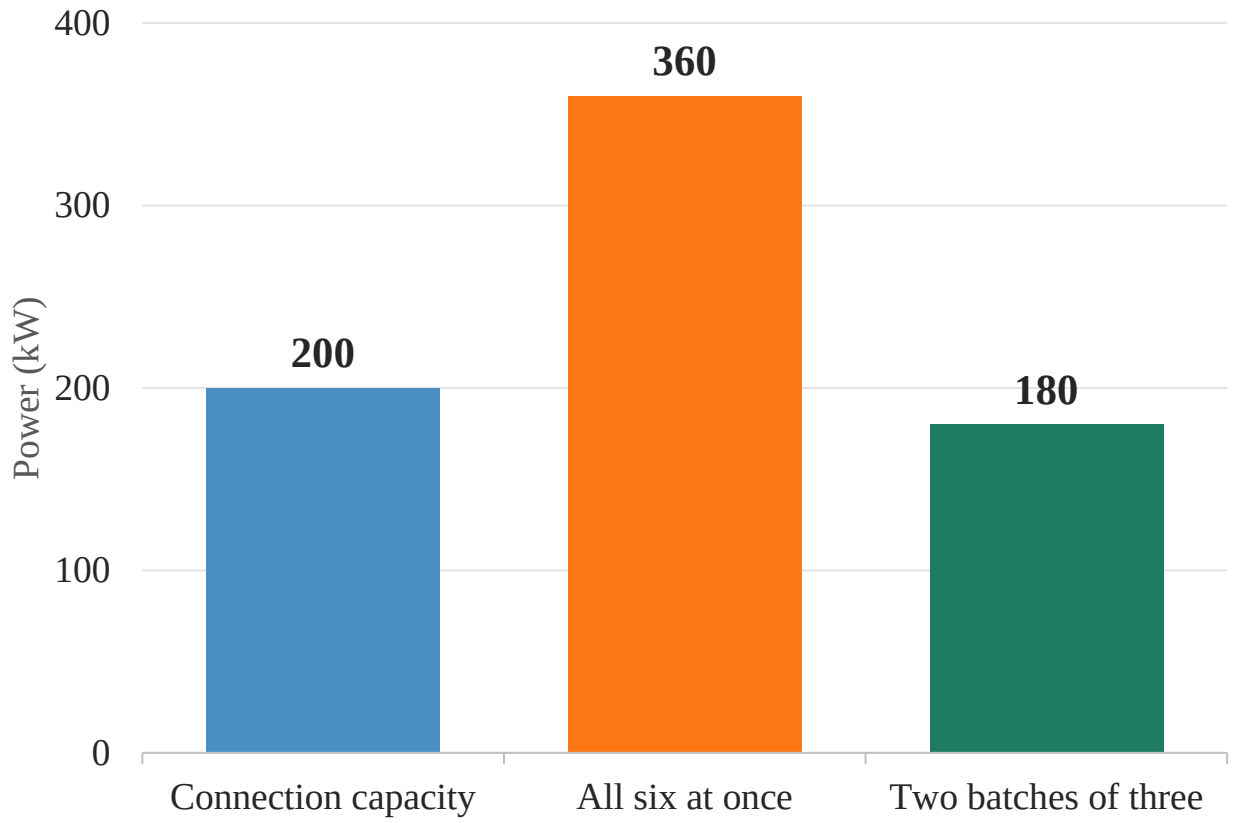
115 [kWh] ÷ 60 [kW] ≈ 1.9 [h].

Units check: [kWh] ÷ [kW] = [h]. The unit of the answer is the one we asked for.

Say the assumption: real charging slows near full and wastes some energy, so the true time is longer.

Method: Kosky et al., Ch. 2 (The Need-Know-How-Solve Method). Example values are illustrative.

Answering the Hook: Six Buses, 200 kW



THE NUMBERS

Energy: 6 × 115 = 690 kWh

Window: 22:00 to 06:00 = 8 h

Average power: 690 ÷ 8 ≈ 86 kW

Two batches: 3 × 60 = 180 kW, utilisation 0.9

Time: 2 × 1.9 h ≈ 3.8 h, inside 8 h

Energy was never the problem. Power was, and staggering the charging solves it.

Illustrative values.

Unit Traps

Slip	Why it is wrong	Say instead
“The charger uses 60 kWh.”	kWh is energy; a charger rating is a rate	It delivers 60 kW, or 60 kWh each hour.
“The bus needs 115.”	No unit and no time basis	115 kWh per bus per day.
“A 0.2 MW connection is 20 kW.”	MW to kW is a factor of 1,000	0.2 MW = 200 kW.
“36 minutes is 0.36 hours.”	Minutes are not decimal hours	36 min ÷ 60 = 0.6 h.

When an answer looks odd, check the units and the time basis first.

Two Handy Flow Rules

RULE 1

Flow = speed × density

Cars passing a point per hour = speed × cars per km.

15 [km/h] × 20 [cars/km] = 300 [cars/h]

RULE 2

Number inside = arrivals × time

People inside = arrival rate × average time inside.

10 [students/min] × 3 [min] = 30 students in the line

Both rules hold for steady conditions. We return to queues in week 9.

Rule 1: Kosky et al., Ch. 8 (highway capacity). Rule 2: standard queueing result. Illustrative values.

Can Ten Buses Charge?

The fleet grows to 10 buses. Same 115 kWh each, 60 kW chargers, 200 kW connection, 8 h window. In pairs, answer:

1. What is the total energy needed?
2. What is the average power over the window?
3. How many chargers can run at once?
4. How many batches are needed?
5. Does everything finish within 8 hours?

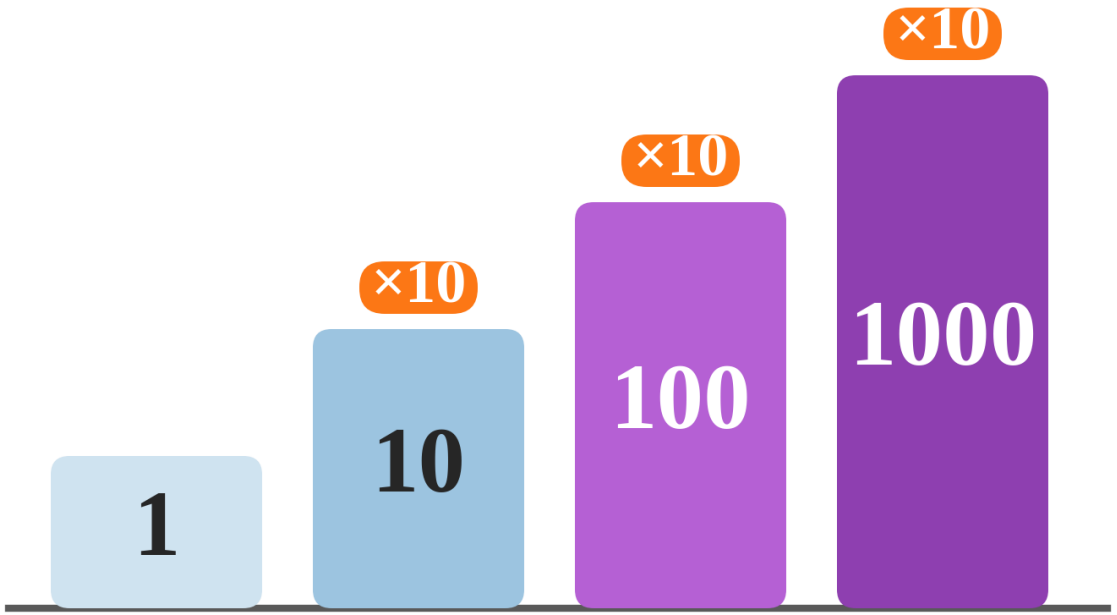
Five minutes. Carry units in square brackets.

Can Ten Buses Charge? Answer Key

	Question	Answer	Working
1	Total energy	1,150 kWh	10 × 115 [kWh]
2	Average power	about 144 kW	1,150 [kWh] ÷ 8 [h] = 143.75 [kW]
3	Chargers at once	3	200 [kW] ÷ 60 [kW] = 3.3, round down
4	Batches	4	10 buses ÷ 3 per batch = 3.3, round up
5	Time needed	7.7 h, just fits	4 × 1.9 [h] ≈ 7.7 [h], below 8 [h]

It fits, but with almost no margin. One slow charger or a late bus would break the plan.

PART 3



each step is a factor of ten

Order of Magnitude

Estimate first, to within a factor of ten.

Estimate Before You Calculate

BEFORE

Know the size to expect

Is the answer tens, hundreds, or thousands?

DURING

Catch slips

A wrong power of ten or a wrong unit shows up at once.

AFTER

Judge the result

Is the answer reasonable compared with what we know?

Kosky et al. estimate the barbershops in a city of 60,000 people at 21. A count in the phone directory gave 23.

An order-of-magnitude estimate is right to within a factor of about 10, and often much closer.

Kosky et al., Ch. 2 (Example 2.4 and problem 10).

Everyday Anchors for Power and Energy

Thing	Typical size	In other units
LED bulb	10 W	0.01 kW
Electric kettle	2 kW	2,000 W
Phone battery	15 Wh	0.015 kWh
City electric bus battery	200 kWh	0.2 MWh
Faculty building at peak (our example)	180 kW	0.18 MW

1 kW = 1,000 W. 1 MW = 1,000 kW. 1 kWh = 1,000 Wh.

Approximate, rounded values. Prefixes: Kosky et al., Ch. 2 (Engineering Units of Measurement).

The Estimation Method

1. **Define** the quantity and its units.
2. **Break it into factors** you can guess.
3. **Guess and round** each factor to one digit, then multiply.
4. **Check** the units, and compare with a known value.

Carry units in square brackets: the units check the method as well as the answer.

Method: Kosky et al., Ch. 2 (Example 2.4).

Worked Example: Solar for One Bus

115 kWh
needed per day
one bus

→

4 kWh
per kW of panels per day
guess, yearly average

→

≈ 30 kW
array size
115 ÷ 4 ≈ 29

→

≈ 150 m²
roof area
30 × 5 [m²/kW]

About 30 kW on about 150 m²: a rooftop, not a field.

Units: [kWh/day] ÷ [kWh/kW-day] = [kW], then [kW] × [m²/kW] = [m²].

Guesses: 4 kWh per kW per day, 5 m² per kW (illustrative). Compare Lecture 5: 5 kW gave about 31 kWh on a clear day.

Estimate It in Your Team

Choose one question. Work through the four steps and write down your assumptions.

A

How many passengers can the campus bus move in one hour?

Hint: seats, trips per hour, standing room.

B

How much electricity does the faculty building use in one day?

Hint: rooms, lamps, computers, hours of use.

1 Define | **2 Break into factors** | **3 Guess and round** | **4 Check units**

Share one sentence: “About X, give or take a factor of 3, assuming …”

Bound Your Estimate

OPTIMISTIC

5 kWh per kW per day

115 ÷ 5 ≈ 23 kW

BEST GUESS

4 kWh per kW per day

115 ÷ 4 ≈ 29 kW

CAUTIOUS

3 kWh per kW per day

115 ÷ 3 ≈ 38 kW

Precise is not the same as accurate: report “about 30 kW”, not “29.17 kW”.

From 23 to 38 kW the decision is the same: a rooftop array. When more precision cannot change the decision, stop refining.

Illustrative values.

Common Pitfalls

Pitfall	Example	Fix
Mixing kW and kWh	“The array makes 5 kWh.”	Power is kW, energy is kWh.
Forgetting the peak	Sizing the connection for the 86 kW average	Compare with the peak.
Double counting	Counting seats per bus and per trip twice	List each factor once.
False precision	“29.17 kW” from rough guesses	One or two digits.

Most estimation errors are unit or counting errors, not hard mathematics.

Key Takeaways

Demand and capacity

What is asked against what can be delivered, with units and a time basis.

Utilisation

Demand ÷ capacity. Near 1 is stressed, above 1 fails.

Rates and flows

Quantity per time. Power × time = energy. Change = in − out.

Size for the peak

Averages hide peaks. Staggering can fix a power problem.

Estimate first

Break into factors, round to one digit, check units.

Bound it

If the decision does not change across the range, stop refining.

Independent Study: Numbers for Your System

1. Write the demand and the capacity of your system, each with a unit and a time basis. Compute the utilisation.

2. Do one rate or flow calculation for your system, carrying units in square brackets.

3. Make one order-of-magnitude estimate for your system, and write down its assumptions.

4. Check each result: the units, the size, and one known value.

Nothing to hand in. This is working material for your team, and it prepares next week: limited and incomplete data.



Thank You — Questions?

COMING UP NEXT

Lecture 7: Quantitative Reasoning II

Reasoning with limited or incomplete data, and comparing interpretations of the same problem.